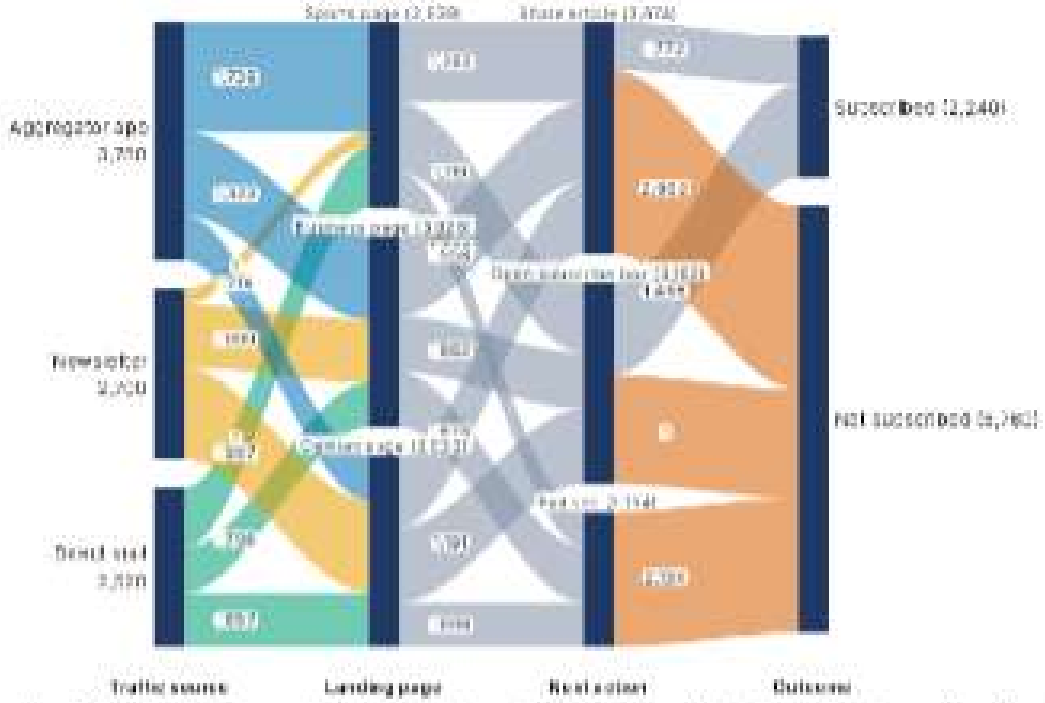
DSA06 - Data Handling and Visualization | Activity 6: Website Flow Sankey Design | Slot C

R.No: 32 Reg. No: 192424278 Name: SHEIK MOHAMMED AFNAN A

Scenario: The analytics team of a news portal tracked 9,000 visits last month through four stages: traffic source, landing page, next action and outcome. The Sankey diagram below shows the number of visits on every flow. Three flows are hidden and marked "?".



(a) Using conservation of flow (what enters a node equals what leaves it) find the three hidden flows, showing working: Newsletter to Sports page; Newsletter to Opinion page; Open subscribe box to Not subscribed. State the rule you used. [4]

(b) Compute: (i) the overall conversion rate, i.e. Subscribed as a percentage of all visits; (ii) the largest single flow into "Not subscribed" or "Exit site"; (iii) the percentage of "Sports page" visits that go on to "Share article". [4]

(c) Design the final Sankey: give four design decisions (node order to reduce crossings, colour rule, labels, what to highlight) and justify each in one line using this data. [4]

(d) The Sankey is to be printed with 1 cm of node height = 500 visits. Compute the height in cm of each landing-page node (Opinion page, Business page, Sports page). [3]

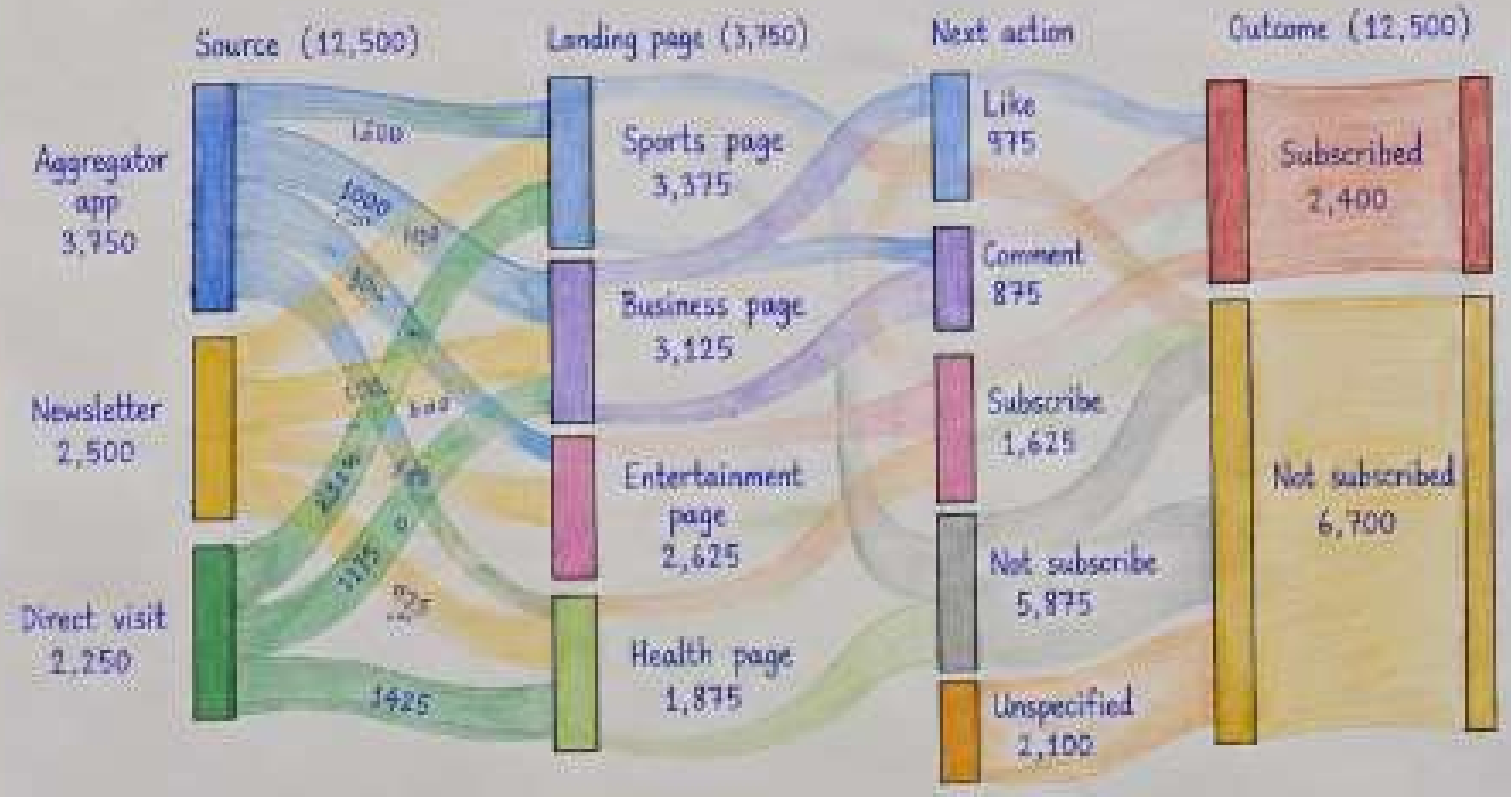
(e) Write R code: a links data frame for the first four flows leaving "Aggregator app" and "Newsletter" in the table order of the diagram, a nodes data frame, and the networkD3::sankeyNetwork() call (or plotly) that draws the whole diagram. [4]

(f) State two situations in which a Sankey diagram misleads or becomes unreadable, and name one alternative chart for each. [2]

DSA06 - Data Handling and Visualization | Activity 6 : Website Flow Sankey Design | Slot C

Reg. No : 192424278	Name : SHEIK MOHAMMED AFNAN A	R No : 32

Scenario : The analytics team of a news portal tracks 9,000 visits across fstops: traffic source, landing page, next action and outcome. The Sankey diagram below shows number of visits on every flow. Three flows are hidden and marked "?".



(d) Node heights (1 cm = 500 visits)

Node	Visits	Height (cm)
Aggregator app	3,750	7.50
Newsletter	2,500	5.00
Direct visit	2,250	4.50
Sports page	3,375	6.75
Business page	3,125	6.25
Entertainment page	2,625	5.25
Health page	1,875	3.75
Like	975	1.95
Comment	875	1.75
Subscribe	1,625	3.25
Not subscribe	5,875	11.75
Unspecified	2,100	4.20
Subscribed	2,400	4.80
Not subscribed	6,700	13.40

(a) Using conservation of flow, find the three hidden flows.

Hidden flow	Value (visits)
Flow to Health page (from traffic sources)	1,875
Next action (Unspecified) from landing pages	2,100
Flow to Subscribed (from "Subscribe")	1,625

(b) Compute the overall conversion rate, subscriber percentage and largest angle.

Overall conversion rate	Subscriber percentage	Largest angle flow (Not subscribed)
$\frac{2,400}{12,500} \times 100 = \boxed{19.2\%}$	$\frac{2,400}{12,500} \times 100 = \boxed{19.2\%}$	$\frac{6,700}{12,500} \times 360 = 192.96^\circ \approx 193^\circ$

(c) Final Sankey diagram (labels, colours and justification)

Stage	Node	Colour	Reason
Traffic source	Aggregator app		External app traffic
	Newsletter		Email subscribers
	Direct visit		Direct users
Landing page	Sports page		Sports content
	Business page		Business content
	Entertainment page		Entertainment content
	Health page		Health content
Next action	Like		Positive / neutral action
	Comment		User engagement
	Subscribe		Key conversion action
	Not subscribe		Did not subscribe
	Unspecified		Other / unknown actions
Outcome	Subscribed		Positive outcome
	Not subscribed		Negative outcome

(e) R code (data frame)

source	target	value
Aggregator app	Sports page	1200
Aggregator app	Business page	1000
Aggregator app	Entertainment page	800
Aggregator app	Health page	450
Newsletter	Sports page	1000
Newsletter	Business page	1200
Newsletter	Entertainment page	300
Newsletter	Health page	0
... (remaining rows	for all flows)	

Create rows for all source → landing page, landing page → next action, next action → outcome flows with the values shown in the diagram.)

(f) Two situations when Sankey is unsuitable and alternatives.

Situation	Alternative chart
1. When the number of nodes / flows is very large and the diagram becomes complex and unreadable.	Alluvial plot or Interactive Sankey with filtering.
2. When precise quantitative comparison between categories is required.	Stacked bar chart or Grouped bar chart